PROJECT OVERVIEW · NOVEMBER 2026

CLONEGUARD

Checks whether a caller's voice matches a trusted enrolled person, and scans the conversation for Hinglish scam patterns. Two layers of evidence, one risk score.

Layer 1 Voiceprint match

Layer 2 Hinglish scam scan

Score 0–100, Low / Caution / High

Built on Gnani AI · Prisma · Voice-clone embeddings · Timbre

#GnaniAI #GreatIndianAIInternshipChallenge

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01 · THE PROBLEM

"BETA, MAIN PAPA BOL RAHA HOON..."

A call comes in. The voice says it is a parent, a boss, or a bank officer. It is urgent: there has been an accident, money is needed now, and nobody else should be told.

Voice cloning and plain human impersonation make this script cheap to run. In India it is often spoken in Hinglish — Hindi written in English letters, mixed with English — so English-only scam filters miss the words that matter.

THE SCRIPT, FROM THE TEST SET

> "Mera accident ho gaya hai... operation ke liye pachpan hazaar rupaye turant chahiye... Paytm par paise bhej do. Kisi ko abhi mat batana."

URGENCY	PRESSURE	ACTION
"turant"	"don't tell anyone"	"send the money"

Two questions decide the call: is this really the person they claim to be, and does the conversation follow a known scam pattern?

02 · WHAT CLONEGUARD IS

A DESKTOP TOOL THAT CHECKS A CALL TWO WAYS

CloneGuard compares the caller's voice against a stored voice fingerprint of the person they claim to be, then reads the transcript for Hinglish scam patterns. The screen shows both pieces of evidence and one combined risk score — and a human makes the final decision.

WHO IT IS FOR

Households and small teams who want to check a suspicious call against someone they trust — without a sales call or a contract.

WHAT IT DOES

Voice identity check + Hinglish scam-phrase scan → a 0–100 score, a band, a plain-English verdict, a spoken verdict and a PDF evidence report.

WHERE IT RUNS

The interface runs locally at `localhost:8501`. The analysis runs on Gnani's cloud APIs, so an internet connection is required.

EIGHT FEATURES

Multi-voiceprint enrollment with claim detection · two-layer composite score · Hinglish scam-call generator · Self-Test Harness · Streamlit UI · incident log · PDF evidence reports · spoken verdict via Timbre

03 · HOW IT WORKS

ONE RECORDING IN, ONE VERDICT OUT

01
Validate
Reject a missing or empty file before any API credit is spent.

→

02
Transcribe
Prisma speech-to-text, `indic-auto`. Cached per file.

→

03
Scan
Layer 2: 32 dual-script patterns over the normalised transcript.

→

04
Resolve claim
"Main Papa bol raha hoon" → compare against Papa's voiceprint.

→

05
Compare
Layer 1: 768-number embedding, cosine vs threshold 0.55.

→

06
Score
Equal-weight composite 0–100 → Low / Caution / High.

EVIDENCE STAYS VISIBLE

Both layer scores, the threshold, every matched phrase with its snippet, and the plain-language sentence are always on screen.

EVERY CHECK IS LOGGED

Timestamp, claimed identity, both layer scores, verdict and transcript excerpt go to the local incident log.

CREDIT-AWARE

Transcripts and embeddings are cached, so re-checking the same unchanged file makes no new API calls.

04 · LAYER 1

VOICEPRINT MATCH

The caller's audio becomes a voiceprint: a list of 768 numbers that describes how a voice sounds (an embedding). It is compared with the enrolled voiceprint of the claimed person using cosine similarity — a 0-to-1 score where 1 means identical voices.

- Score at or above `0.55` → identity match
- Score below `0.55` → identity mismatch
- Several voiceprints can be enrolled: Papa, Mummy, CEO…
- Re-enrolling the same role replaces that voiceprint

LAYER 1 RISK

```
layer1_risk = clamp((threshold - cosine) / threshold, 0, 1)
# cosine 0.2752 → (0.55 - 0.2752) / 0.55 = 0.50
# cosine ≥ 0.55 → 0.00
```

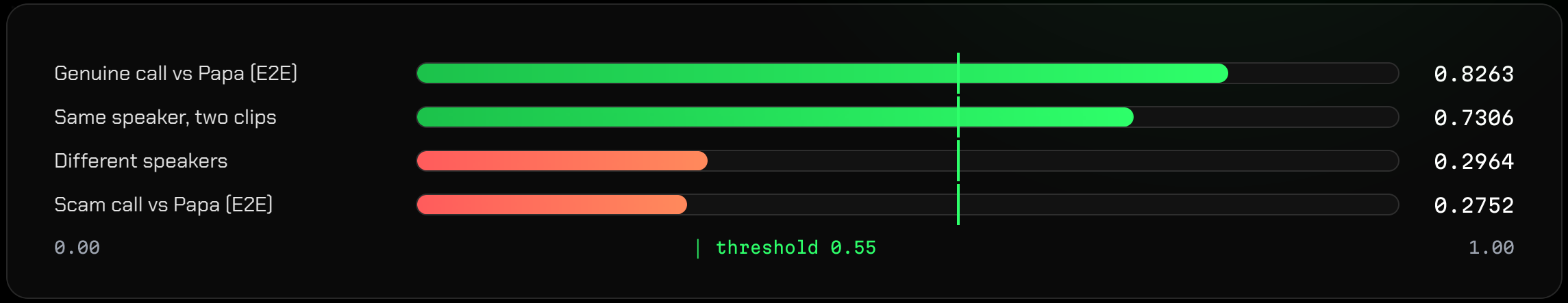
WHAT A MISMATCH MEANS

Layer 1 flags an identity mismatch between the caller and the enrolled voice. That catches cloned voices and human impersonators alike — it is not, on its own, proof that the audio was machine-generated.

05 · MEASURED

CAN IT TELL VOICES APART?

Cosine similarity on the project's recordings, measured with `voiceprint_test.py` and the end-to-end run. The fingerprint step is deterministic: the same audio always produces the same numbers.



SEPARATION GAP

0.4342

FINGERPRINT SIZE

768 numbers

THRESHOLD

0.55

06 · LAYER 2

HINGLISH SCAM-PATTERN SCAN

The transcript is normalised (lowercase, punctuation removed, whitespace collapsed) and scanned for 32 patterns with 204 variants, each stored in both Latin and Devanagari script — so the same rules work whichever script Prisma returns.

URGENCY · 5

तुरंत · abhi karna hoga

"right now", "must do it now"

OTP / BANK · 5

OTP batao · account number

requests for codes and details

AUTHORITY · 5

police · bank officer · CEO

impersonating someone in charge

THREATS · 6

case darj · account band

arrest, blocking, legal action

EMOTIONAL · 5

एक्सीडेंट हो गया है

accident, hospital, "don't tell anyone"

CALL TO ACTION · 6

पैसे भेज दो

"send the money", payment links

```
layer2_risk = min(1, Σ hit weights / 0.8)  # four 0.20 hits → 0.8 / 0.8 = 1.00
```

07 · CLAIMED IDENTITY

WHO DOES THE CALLER SAY THEY ARE?

The transcript is searched for identity claims. A claim selects that person's voiceprint, so the caller is compared with the right enrolled voice instead of a generic one.

CLAIM DETECTED · ENROLLED

> "Main Papa bol raha hoon"

→ compare against Papa's voiceprint. Layer 1 is scored.

CLAIM DETECTED · NOT ENROLLED

> "Main CEO bol raha hoon"

→ on-screen note "claim names someone not enrolled"; the score comes from Layer 2 alone, rescaled to 0–100.

NO CLAIM

> "Hi, dinner at eight?"

→ compare against the default voiceprint, and say so on screen.

BUILT-IN CLAIM LOOKUPS

- **Papa** papa bol raha hoon · पापा बोल रहा हूँ · beta main hoon
- **Mummy** mummy bol rahi hoon · मम्मी बोल रही हूँ
- **CEO** ceo bol raha hoon · boss bol raha hoon · सीईओ

08 · THE COMPOSITE SCORE

TWO RISKS, ONE NUMBER

```
composite = round(100 × (0.5 × layer1_risk + 0.5 × layer2_risk))

# scam call: 100 × (0.5 × 0.50 + 0.5 × 1.00)
# = 100 × 0.75 = 75
```

Weights stayed at 0.5 / 0.5: calibration found no evidence for moving them. The arithmetic is shown on screen under "How this score was built".

0–29 · LOW RISK

"The caller's voice is consistent with the enrolled person and no scam patterns matched. This is a weak signal, not a clearance."

30–59 · CAUTION

"One layer raised a concern. Ask a verification question the real person would answer."

60–100 · HIGH RISK

"The caller does not match the claimed person's voice, or the conversation uses known scam patterns, or both. Treat this call as unsafe."

09 · BUILT ON GNANI AI

THREE GNANI SERVICES, ONE VERDICT

PRISMA · SPEECH TO TEXT

prisma-v3.0

Transcribes the call with `indic-auto` language detection. Returns Hindi in Devanagari or Latin script — the pattern list covers both.

VOICE-CLONE · EMBEDDINGS

768-d

Turns a recording into a voice fingerprint. The vector arrives as a string, so `voiceprint.py` parses JSON, CSV, tensor-repr and base64 defensively.

TIMBRE · TEXT TO SPEECH

timbre-v2.5

Speaks the verdict aloud (voices Nalini / Deepak) and voices the built-in Hinglish scam and genuine test calls.

COST PER CHECK

About 2 API calls for a new recording (transcript + embedding), 0 for a cached one, 1 to enroll a new voice and 1 to speak a verdict. The API key is read from the `GNANI_API_KEY` environment variable and never written to disk.

10 · ARCHITECTURE

SMALL MODULES, ONE JOB EACH

```
app.py — Streamlit UI, five pages
config.py — thresholds, weights, paths
gnani.py — Prisma + Timbre client
voiceprint.py — embeddings + cosine
engine/
  pipeline.py — run_check()
  layer1.py — enroll / compare
  layer2.py — normalise / scan
  claim.py — identity claims
  score.py — composite + bands
generator/ — Hinglish call scripts + TTS
selftest/ — harness + scorecard PDF
reports/ — per-check evidence PDF
```

LOCAL

Streamlit interface on `localhost:8501`, JSON storage, caches, incident log and PDF reports — all on the user's machine.

CLOUD

Inference only: Prisma, voice-clone embeddings and Timbre on `api.vachana.ai` over HTTPS.

STACK

Python · Streamlit 1.66 · fpdf2 + HarfBuzz (Devanagari PDFs) · pypdf · requests

11 · DATA & STORAGE

PLAIN JSON FILES, KEPT LOCAL

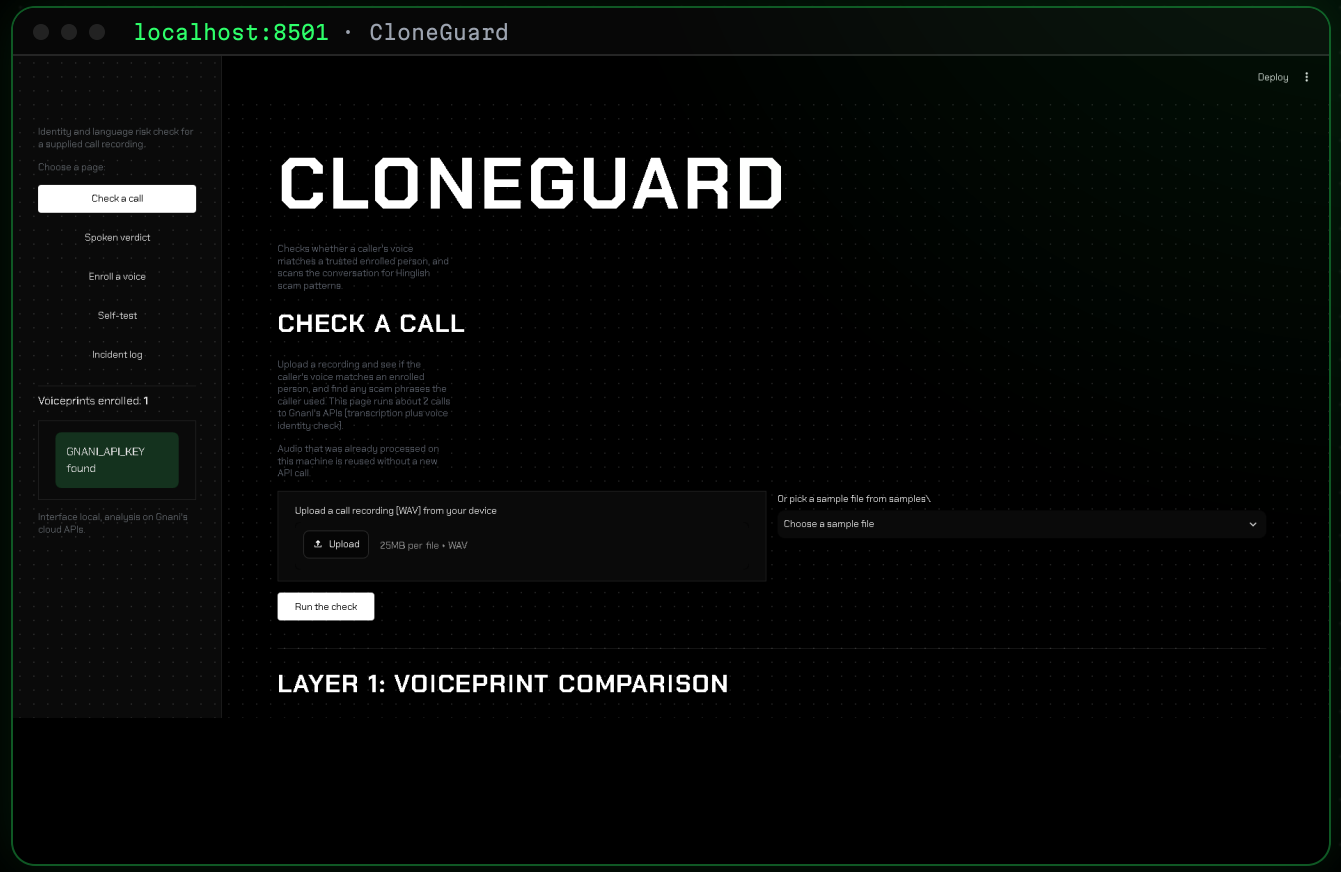
FILE	WHAT IT HOLDS	IN GIT?
data/voiceprints.json	id, name, role, 768-number embedding, source, created_at	No (gitignored)
data/incidents.json	Every check: timestamp, claim, both layers, composite, band, verdict, transcript	No (gitignored)
data/stt_cache.json	Prisma transcripts keyed by file path, size and modified time	No (gitignored)
data/embedding_cache.json	Embeddings keyed the same way, so unchanged audio is never re-billed	No (gitignored)
data/patterns.json	32 Layer 2 patterns, 204 Latin + Devanagari variants, weights	Yes
data/scenarios.json	10 labeled test calls: script, voice, audio path, expected label	Yes

DESIGN CHOICE

The record counts stay small, so a database was not needed. Writes go to a temp file and are atomically replaced, so a crash never leaves a half-written log. No audio is stored in the incident log.

12 · THE INTERFACE

FIVE PAGES, ONE SIDEBAR



CHECK A CALL

Upload a WAV or pick a sample; see both layers and the score.

SPOKEN VERDICT

Hear the verdict read by Timbre.

ENROLL A VOICE

Add, replace or remove trusted voiceprints.

SELF-TEST

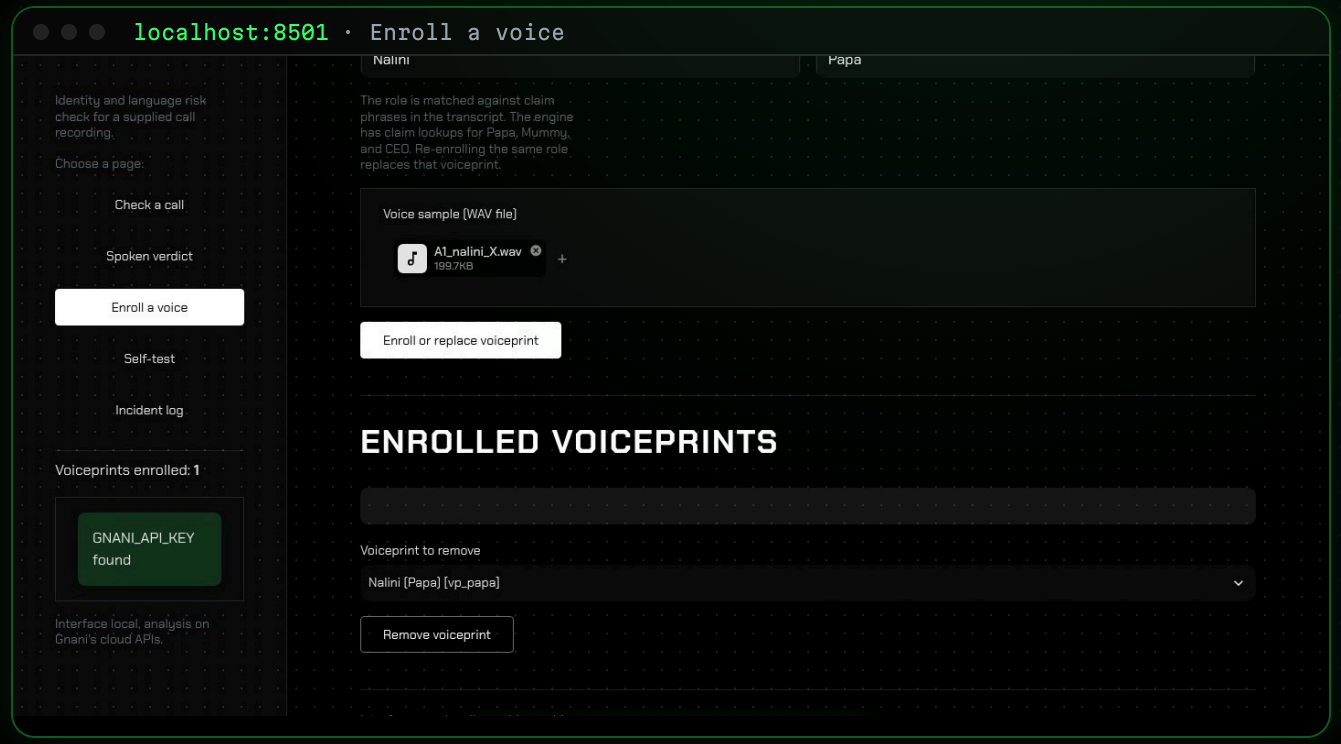
Run the battery; export the scorecard PDF.

INCIDENT LOG

Every check, filterable by date.

13 · WALKTHROUGH 1

ENROLL A TRUSTED VOICE



Type a name and a role, upload a short WAV, press Enroll or replace voiceprint.

IN THE RECORDED TEST

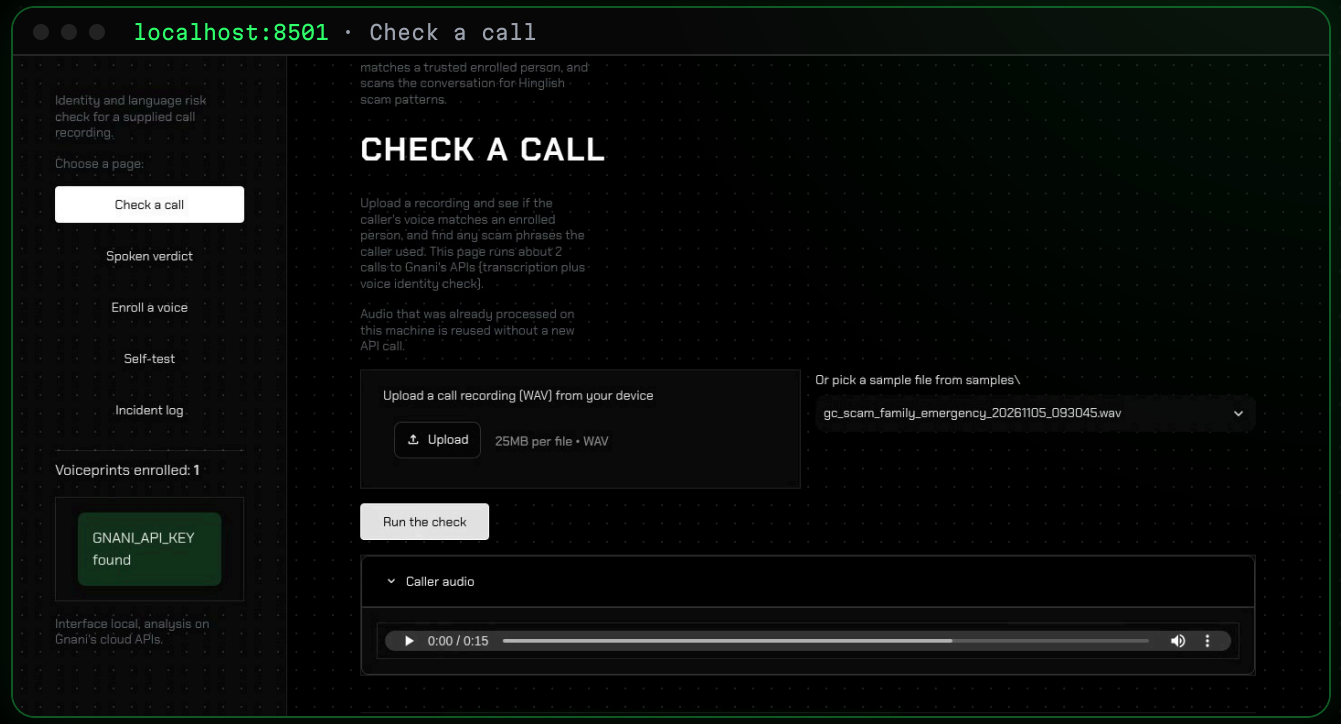
Name "Nalini", role "Papa", sample `A1_nalini_X.wav` → voiceprint `vp_papa`, one embedding call.

WHY THE ROLE MATTERS

The role is matched against claim phrases in the transcript. When a caller says "main Papa bol raha hoon", this is the voiceprint they are compared with.

14 · WALKTHROUGH 2

A SCAM CALL COMES IN



The test call `gc_scam_family_emergency.wav` claims to be Papa — but it is voiced by a different Timbre voice (Deepak).

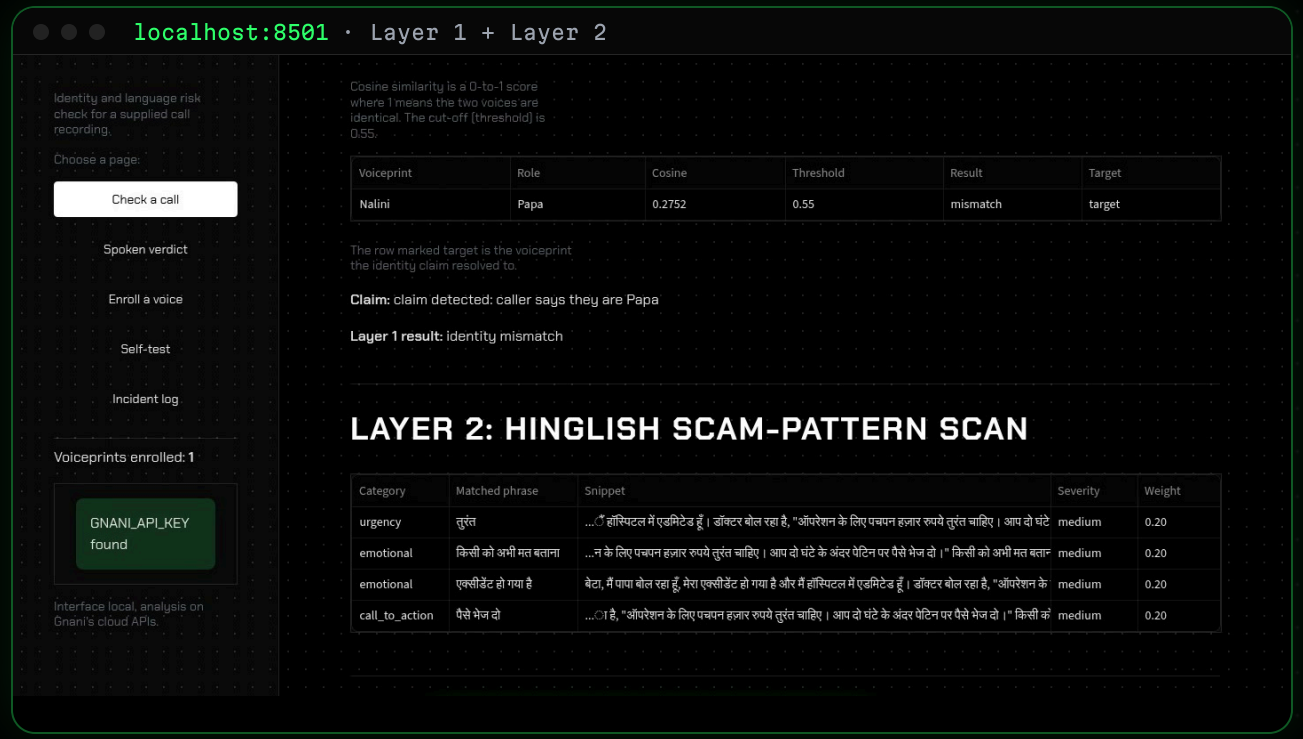
WHAT THE CALLER SAYS

> "Beta, main Papa bol raha hoon. Mera accident ho gaya hai... pachpan hazaar rupaye turant chahiye... Paytm par paise bhej do. Kisi ko abhi mat batana."

Press Run the check: about two Gnani calls for a new file, none for a cached one.

15 · WALKTHROUGH 3

THE RESULT: 75 / 100 · HIGH RISK



LAYER 1 · VOICE

0.2752 vs threshold 0.55 → identity mismatch.
Claim detected: caller says they are Papa.

LAYER 2 · 4 RED FLAGS

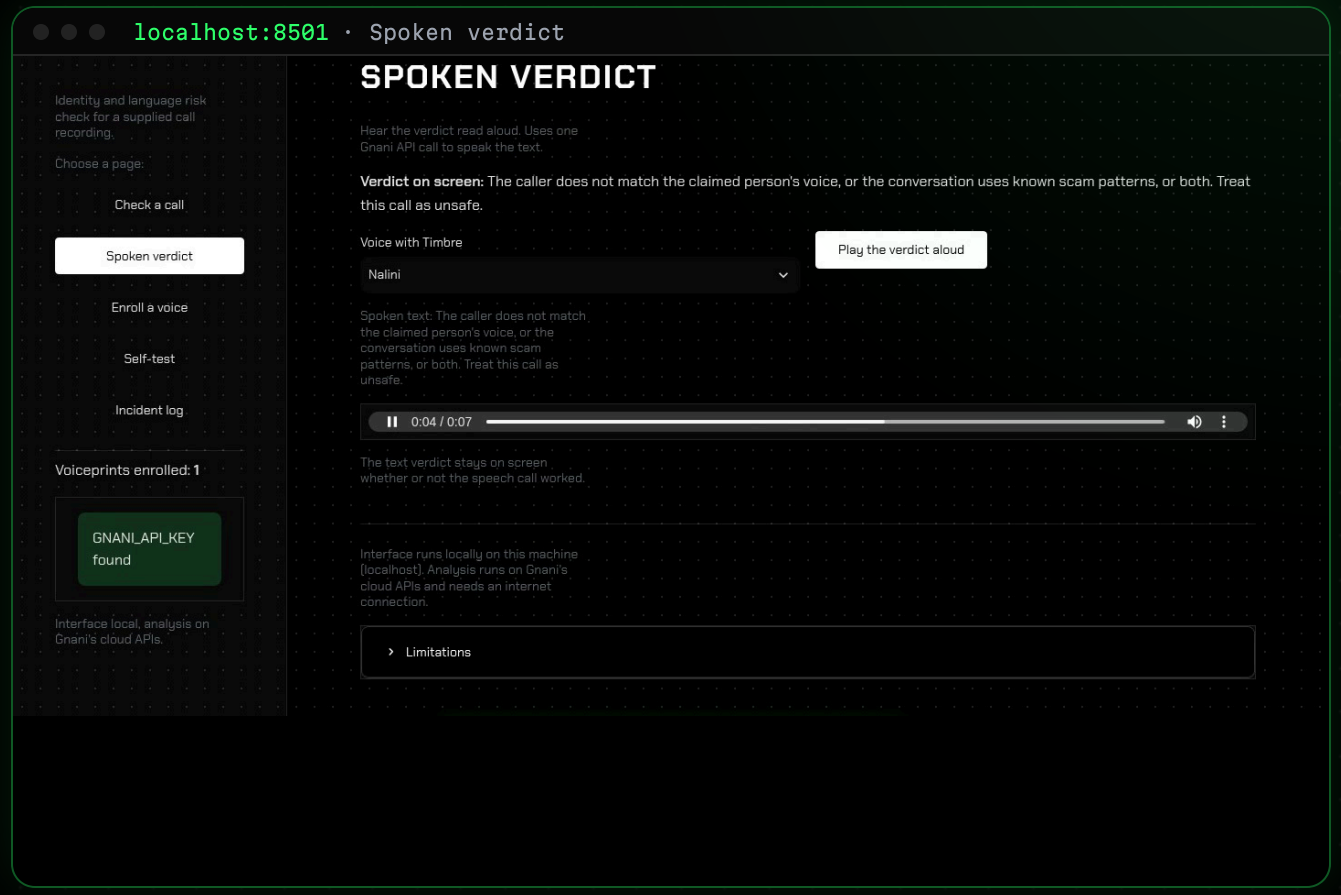
तुरंत · एक्सीडेंट हो गया है · किसी को अभी मत बताना · पैसे भेज दो

urgency · emotional · emotional · call to action

75/100 HIGH RISK

16 · WALKTHROUGH 4

SPOKEN VERDICT & EVIDENCE REPORT



SPOKEN VERDICT · TIMBRE

One click reads the on-screen verdict aloud in a selectable voice. The text stays on screen whether or not the speech call worked.

PDF EVIDENCE REPORT

One A4 PDF per check: both layer scores, matched phrases, transcript excerpt with native Devanagari, verdict, and the four known limitations. Built from the stored record — no new API calls.

INCIDENT LOG

Every check is listed with timestamp, claimed identity, both layer scores, score, band and transcript excerpt.

17 · SCAM-CALL GENERATOR

TEST CALLS WITH KNOWN ANSWERS

The build machine had no microphone, so every test file is generated: Hinglish scripts spoken by Timbre voices, with a phone-line noise effect. Each one carries its true label.

SCENARIO	TYPE	VOICE	NOTE
Family emergency	scam	Deepak	"Main Papa bol raha hoon" — claim routing
Bank fraud · Government / police · Tech support · Prize refund	scam	Deepak	OTP, authority, threats, refunds
Executive	scam	Deepak	"Main CEO bol raha hoon"
Bank refund, no urgency	scam	Deepak	Hard case: a scam without urgency words
Friend dinner · Relative check-in	genuine	Nalini	Ordinary conversation
College exam fee	genuine	Nalini	Hard case: genuine call that mentions money

18 · SELF-TEST HARNESS

THE TOOL ATTACKS ITSELF

One button runs every labeled scenario through the full pipeline and prints a calibration scorecard. A scam only counts as caught at High; a genuine call only passes at Low; Caution is a miss in both directions.

SCAM CAUGHT	GENUINE PASSED	CAUTION	ERRORS
7/7	3/3	0	0

WEAKEST SCENARIO

Named on every scorecard, with its distance from the expected band — so the next tuning target is visible.

HONEST FRAMING

"Calibration against builder-written scripts, not independent accuracy." — printed on the scorecard itself.

19 · CALIBRATION RESULTS

PER SCENARIO, AND ACROSS THRESHOLDS

#	SCENARIO	EXPECTED	SCORE	BAND
1	family_emergency	scam	75	high
2	bank_fraud	scam	74	high
3	executive	scam	75	high
4	government_police	scam	76	high
5	tech_support	scam	74	high
6	prize_refund	scam	76	high
7	friend_dinner	genuine	0	low
8	college_exam_fee	genuine	0	low
9	relative_checkin	genuine	0	low
10	bank_refund_no_urgency	scam	71	high

THRESHOLD	SCAM	GENUINE	CAUTION
0.45	7/7	3/3	0
0.50	7/7	3/3	0
0.55	7/7	3/3	0
0.60	7/7	3/3	0
0.65	7/7	3/3	0

READING THE SWEEP

All five thresholds return the same result, so 0.55 is not a knife-edge fit on this set. Reproduced on a fresh machine: run `run_20261110_082347_ef50`.

20 · FRESH-MACHINE END-TO-END TEST

CLEAN CLONE, LINUX, REAL API CALLS

PHASE	CHECK	RESULT
Preflight	Key present, endpoint reachable (422 probe, no credit)	PASS
Static	21 Python files compile · wording scan · palette scan (0 illegal colours)	PASS
UI smoke	5 pages, 0 exceptions, HTTP 200 on localhost	PASS
Enroll	Nalini as Papa → voiceprint stored	PASS
Scam check	Claim → Papa · cosine 0.2752 · 4 red flags · 75 · High	PASS
Genuine check	Cosine 0.8263 · 0 flags · 0 · Low	PASS
Claim routing	"CEO" not enrolled → note shown; then CEO enrolled → routed	PASS
Reports	Evidence PDF + scorecard PDF readable · spoken verdict returned	PASS
Self-test	7/7 · 3/3 · caution 0 · errors 0 · sweep flat 0.45–0.65	PASS

Whole session: 27 Gnani API calls. Recorded as a full 1440×900 video with the app's own audio.

21 · PRIOR ART & POSITIONING

WHAT EXISTS, AND WHERE THIS FITS

EXISTING WORK	WHAT IT DOES	CLONEGUARD'S DISTINCTION
Pindrop	Enterprise identity trust for contact centres: synthetic-speech detection, caller authentication, fraud scoring.	Consumer desktop workflow; no sales contract.
Resemble AI	Multimodal deepfake detection and an Identity API (beta) that matches audio to enrolled voices.	Hinglish scam scan alongside identity, with visible arithmetic.
Tring	Indian AI call assistant: Hinglish, spam blocking, transcripts, answers in a cloned voice.	Verifies a claimed identity instead of creating a voice.
VoxDetect · SIH26104 projects	Open prototypes combining deepfake classifiers, voiceprints and call context.	Built-in scam-call generator, self-test scorecard and per-check PDF.

A partial search cannot prove that any feature is absent from the market; these are feature-level differences from the sources reviewed on 4 Nov 2026.

22 · SECURITY & LIMITATIONS

WHAT THE EVIDENCE SUPPORTS

SECURITY AUDIT · 9 NOV 2026

- [ ] Generated data files removed from git
- [ ] Upload size capped at 25 MB, WAV header checked
- [ ] Sample paths guarded with basename
- [ ] Pinned dependency versions
- [ ] No eval / pickle / exec, no secrets in history

`Local-only breachability: LOW`

1 · A PERFECT CLONE PASSES LAYER 1

Matching an identity is not proof that the audio is genuine.

2 · A MISMATCH IS NOT PROOF OF SYNTHETIC AUDIO

It flags an identity mismatch — clones and human impersonators alike.

3 · ANALYSIS RUNS ON GNANI'S CLOUD

Audio leaves the machine; no internet, no analysis.

4 · A PARTIAL SEARCH CANNOT PROVE ABSENCE

"No patterns matched" never means the call is safe.

23 · WHAT'S NEXT

WHERE IT GOES FROM HERE

01

Background monitor

A small process that watches audio the user supplies and raises the same two-layer signal without reopening the app.

02

Public benchmarks

Evaluate on ASVspoof-style sets and IndicSynth instead of self-generated samples.

03

Measured weights

Tune Layer 2 weights and the identity cut-off from Self-Test data rather than starting values.

CLONEGUARD

Interface on localhost · analysis on Gnani's cloud APIs · Prisma · Voice-clone embeddings · Timbre

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